HY-TECH DRILLING

DRILLER'S ASSESSMENT

(ÉVALUATION DU FOREUR)

Employee Name:

(Nom de l'employée): _____

Date: _____

F0-216

Hy-Tech Drilling Training and Self-Assessment Handbook

(Hy-Tech Drilling Manuel de Formation et d'Auto-Évaluation)

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Introduction

Please review the following information before you begin working through the Hy-Tech Drilling Driller's Assessment. Hy-Tech believes in promoting a culture that values safety, training, and mentoring. We want all employees to develop to be the most capable and professional workers in the industry.

As a result of the time and effort that you've dedicated to your career development with Hy-Tech, we would like you to complete a Diamond Driller assessment.

Driller Assessment

The Driller's Assessment will document your knowledge of processes, equipment, Worker responsibilities and Supervisor responsibilities. An assessor will then review and provide feedback on development opportunities. If there are areas that require further development, the assessor will provide you with feedback based on your answers.

Job Hazard Analysis

You are also required to fill out a Job Hazard Analysis (JHA) in its entirety. There is a fillable JHA attached for you to use. You should select (from memory) a previous JHA that you completed or participated in. The assessor will be looking for an unusual or high-risk task that is not considered part of everyday activities.

Submitting

Please take your time answering these questions. We expect that you should have completed and returned the Driller Assessment & JHA within 2 weeks of receiving it.

To submit, email to: **training@hy-techdrilling.com** or leave it at one of our office locations

If you have any questions, please contact:

Wesley Keating
wesley.keating@hy-techdrilling.com
819-290-1991

Yoann Bellehumeur
yoann.bellehumeur@hy-techdrilling.com
819-290-1991

Lyndon Werbiski
lyndon.werbiski@hy-techdrilling.com
780-297-3166

Kyle Boegel
kyle.boegel@hy-techdrilling.com
250-550-6884

Keegan Pacey
keegan.pacey@hy-techdrilling.com
705-205-0891

Cameron Turner
cameron.turner@hy-techdrilling.com
782-234-0325

Development / Next Steps

Once the assessor has completed their review and provided feedback, they will advise the management group if more development time is needed or if you are ready for placement/mentorship.

Introduction

Veuillez consulter les informations suivantes avant de commencer à travailler sur l'évaluation du foreur Hy-Tech Drilling. Hy-Tech croit en la promotion d'une culture qui valorise la sécurité, la formation et le mentorat. Nous voulons que tous les employés deviennent leurs travailleurs les plus compétents et les plus professionnels de l'industrie.

Grâce au temps et aux efforts que vous avez consacrés à votre développement de carrière chez Hy-Tech, nous pensons que vous êtes sur le point de commencer à vous préparer à devenir un foreur au diamant.

Evaluation du Foreur

L'évaluation du foreur documentera votre connaissance des processus, de l'équipement, des responsabilités des travailleurs et des responsabilités du superviseur. Un évaluateur examinera ensuite et fournira des commentaires sur les opportunités de développement. S'il y a des domaines qui nécessitent un développement supplémentaire, l'évaluateur vous fournira des commentaires en fonction de vos réponses.

Analyse des Risques (Job Hazard Assessment)

Vous devez également remplir une analyse des risques professionnels (JHA) dans son intégralité. Il y a un JHA remplissable ci-joint que vous pouvez utiliser. Vous devez sélectionner (de mémoire) une JHA précédente que vous avez terminée ou à laquelle vous avez participé. L'évaluateur recherchera une tâche inhabituelle ou à haut risque qui n'est pas considérée comme faisant partie des activités quotidiennes.

Soumission

Veuillez prendre votre temps pour répondre à ces questions à choix multiples. Nous nous attendons à ce que vous ayez rempli et renvoyé l'évaluation du foreur et JHA dans les 2 semaines suivant sa réception.

Pour soumettre, envoyez un courriel à : training@hy-techdrilling.com ou laissez-le dans l'un de nos bureaux
Si vous avez des questions, veuillez contacter :

Pat Bourgeois
Patrick.Bourgeois@hy-techdrilling.com
613-293-8914

Yoann Bellehumeur
Yoann.Bellehumeur@hy-techdrilling.com
819-290-1991

Al Gauthier
Al.Gauthier@hy-techdrilling.com
250-643-8040

Développement / Prochaines étapes

Une fois que l'évaluateur a terminé son examen et fourni ses commentaires, il avisera le groupe de gestion si plus de temps de développement est nécessaire ou si vous êtes prêt pour le placement/mentorat.

Maintenance

1. **What do you inspect prior to installing a set of jaws?**

- [ ] a. Bowl is free from debris and wear
- [ ] b. Shims are in place and clean
- [ ] c. Springs are in 2 pieces
- [ ] d. Oil level is all the way to the full mark

2. **How many bolts are used to attach the chuck plate and what are their sizes?**

- [ ] a. 3 x 15/16" bolts and 3 x 1 1/8" bolts
- [ ] b. 3 x 1/2" bolts and 3 x 3/4" bolts
- [ ] c. 3 x 5/8" bolts and 3 x 3/4" bolts
- [ ] d. 3 x 7/16" bolts and 3 x 5/8" bolts

3. **Have you changed, observed or helped change chuck jaws in mid pull? Select the best answer to explain the process observed.**

- [ ] a. I haven't changed, observed or helped change chuck jaws in mid pull.
- [ ] b. Break joint, open chuck, float chuck down past joint, remove rod, slow feed chuck above joint with enough space to give you room, remove plate and replace jaws.
- [ ] c. Open foot clamp, open chuck, remove plate and replace jaws, install tap on rod and go fish rods out of hole.
- [ ] d. This can't be done while pulling rods.
- [ ] e. Break joint, open chuck, float chuck down past joint, remove rod, slow feed chuck above joint with enough space to give you room, close chuck, remove plate and replace jaws.

4. **Did you change, observe or help change foot clamp jaws in mid pull? If yes, explain the process observed.**

5. **What are a few things you should check for if the drill engine is lagging or not running properly?**

- [ ] a. The coolant level is adequate
- [ ] b. The helper filled the drill full with gasoline
- [ ] c. If the air filter is plugged
- [ ] d. That water is flowing through the cooler

6. **If the drill will not start after a few attempts, what would you inspect before calling the foreman?**

- [ ] a. It's been given enough starting fluid
- [ ] b. It's hooked up properly to 24 volts
- [ ] c. The e-stops are pulled out
- [ ] d. The radiator is clear of debris and gunk

7. How do you bleed the fuel lines on the Tech 5000?

- [ ] a. Use the hand pump on the Racor fuel filter, once the ball is hard, crack bleeder screws to let air out until just fuel comes out
- [ ] b. Turn the key on and let the fuel push the air out
- [ ] c. Crank the motor forcing the air out until it fires
- [ ] d. Crack the injection lines and turn over the motor until fuel pulses out

8. Bean pump cups and sleeves have a series of gaskets. Where are they located?

- [ ] a. The crankcase, head and caps
- [ ] b. The rear cover, caps and cups
- [ ] c. The crank, pistons and connecting rods
- [ ] d. The caps and plugs

9. Choose the answer that explains the steps needed to grease a Gorman-Rupp pump (mud mixer pump).

- [ ] a. Turn cross arm counterclockwise to raise arm to top, fill until grease comes out relief hole, turn arm clockwise to return arm to top of plunger.
- [ ] b. The Gorman doesn't need grease, it's a self-lubricating system.
- [ ] c. Pump grease into nipple until grease comes out the relief hole.
- [ ] d. Turn cross arm clockwise to raise arm to top, fill until grease comes out relief hole, turn arm counterclockwise to return arm to top of plunger.

10. When the ignition switch on the drill and supply pump sticks in the "on" position, what will you damage and why?

- [ ] a. The pump - It pressurizes fuel to the injectors and burns out the pump
- [ ] b. The solenoid and flywheel
- [ ] c. The Bendix, solenoid and flywheel
- [ ] d. The injection Pump

Why?

11. What should you check for on a supply or pressure pump if the water slows or if the pump is jacking?

- [ ] a. The crankcase and suction
- [ ] b. The balls and seats
- [ ] c. The cylinders and cups
- [ ] d. The suction and seats

12. Explain the proper procedure for draining the coil stove?

- [ ] a. Turn off coil stove, disconnect hoses and let water drain out
- [ ] b. Disconnect hoses, turn on coil stove, let it steam out 3-4 puffs, turn off coil stove and wait 15 minutes before going inside
- [ ] c. Disconnect hoses, turn on coil stove, let it steam, turn off coil stove and blow in pipe to see if water is all gone
- [ ] d. Disconnect hoses, turn on coil stove, let it steam 2 puffs and wait 5 minutes before going inside

13. **What oil does Hy-Tech use in the supply pump transmission?**

- [ ] a. ATF
- [ ] b. JD Hy-Gard transmission fluid
- [ ] c. 80w-90
- [ ] d. 15w-40

14. **What oil is used in the B-20 Head?**

- [ ] a. ATF
- [ ] b. 80w-90
- [ ] c. 68w-220
- [ ] d. JD Hy-Gard

Drilling

1. **How far back should the lifter case be from the back of the crown? What are the advantages of having the base set at the appropriate measurement?**

- [ ] a. 1/4" and to not grind core
- [ ] b. 1/16" so that it will indicate when it's locked by cutting water off
- [ ] c. 3/16" so that the core breaks properly
- [ ] d. 1/8" so that it will indicate when it's locked by cutting water off

2. **When changing the bit or pulling rods, what should you inspect before re-entering the hole?**

- [ ] a. The locking coupler and adapter
- [ ] b. The reaming shell
- [ ] c. The landing ring
- [ ] d. All downhole equipment

3. **Select the list of 4 things that can happen if you push a block.**

- [ ] a. Make more money, more production, polishes bit, gets the rock nice and tight so you don't drop any
- [ ] b. Push the core spring up the tube, grind core, miss latch the tube, damage the landing body
- [ ] c. Premature wear to the bit, damage the landing ring, damage backend bearings, split lifter case
- [ ] d. Premature wear on bit, damage rubbers, split lifter case, grind core
- [ ] e. Aggravates your helper x 4

4. **Select the explanation of the procedure for installing the casing clamp.**

- [ ] a. Set pull down to 1200 PSI, open the rod clamp and slowly open the feed valve until hold back is 0 PSI and then close clamp
- [ ] b. Set pull down to 1000 PSI, open the rod clamp and slowly open the feed valve until hold back is 0 PSI and then close clamp
- [ ] c. Set pull down to 1200 PSI, open the rod clamp and slowly open the feed valve until hold back is 10 PSI and then close clamp
- [ ] d. Set pull down to 1200 PSI, open the rod clamp and slowly open the feed valve until hold back is 0 PSI and then open the clamp

5. **When you are drilling casing and then experience water loss/dry hole, why is it important to close off the valve to the water swivel until it shows 100 PSI?**

- [ ] a. To keep pressure in hole
- [ ] b. To keep pressure in pump
- [ ] c. To indicate how much water your using
- [ ] d. To indicate that water is going down hole and keep water from siphoning from mud tank

6. **At what PSI should the rod cutter blades (open) and (cut)?**

- [ ] a. Open at 250 psia, cut at 400-600 psia
- [ ] b. Open at 300 psi, cut at 400-600 psi
- [ ] c. Open at 350 psia, cut at 600-800 psia
- [ ] d. Open at 250 psi, cut at 400-600 psi

7. **How do you adjust the power float for breaking out rods?**

- [ ] a. Engage power float lever towards you, turn the pressure reducing cartridge until the head stops moving up, push up on the head by hand, the head should move with you pushing on it, lock the pressure reducing cartridge.
- [ ] b. Engage power float lever away from you, turn the pressure reducing cartridge until the head stops moving up, push up on the head by hand, the head should move with you pushing on it, lock the pressure reducing cartridge.
- [ ] c. Engage power float lever away from you, turn the pressure reducing cartridge until the head moves down, push up on the head by hand, the head should move with you pushing on it.
- [ ] d. Engage power float lever towards you, turn the pressure reducing cartridge until the head stops moving up, lock the pressure reducing cartridge.

8. **How do you anchor the drill to a fly pad (2 processes)?**

- [ ] a. Deadman to the H frame and burn in rod 15° steeper if angle 45°-80° or 15° shallower if angle is 80°-90° and attach monkey
- [ ] b. Deadman to the H frame and burn in rod 10° steeper if angle 45°-80° or 10° shallower if angle is 80°-90° and attach monkey
- [ ] c. Attach saddle anchor to the frame and burn in rod 10° steeper if angle 45°-80° or 10° shallower if angle is 80°-90° and attach clam
- [ ] d. Attach saddle anchor to the frame and burn in rod 15° steeper if angle 45°-80° or 15° shallower if angle is 80°-90° and attach clam

9. **Choose the sequence required when loosening or tightening stiff legs.**

- [ ] a. Loosen far one first and near one last, tighten in reverse
- [ ] b. Loosen near one first and far one last, tighten in reverse
- [ ] c. Loosen far one first and near one last, tighten the same way
- [ ] d. Loosen near one first and far one last, tighten the same way

10. **Select the proper procedure for shifting gears on a B-20 head.**

- [ ] a. Take pin out, rotate chuck with Hyd. Power slowly and pull down shift lever
- [ ] b. Take pin out, use wrench on bolt head to rotate chuck while pulling on shifter until it meshes into position
- [ ] c. Turn lockout lever into safe position, take pin out, tap handle down with a rubber mallet gently until it shifts into position
- [ ] d. Turn lockout lever into safe position, take pin out, use wrench on nut to rotate chuck while pulling on shifter until it meshes into position

11. What can or will happen if you use machine power when trying to shift gears on a B-20 head?

- [ ] a. It will damage the flux capacitor inside the head
- [ ] b. The Proton accelerator will need recalibration
- [ ] c. It will damage the internal gears
- [ ] d. It will shift smoother and better

12. If the locking coupling does not have a tang or if it has been cut off, which parts will wear out?

- [ ] a. The landing ring
- [ ] b. The tube and lifter case
- [ ] c. The latch ears
- [ ] d. The locking coupler adapter

13. At what Depth Should the Assist Ram be installed?

- [ ] a. 300-500mHQ or 600-800mNQ
- [ ] b. 450-500mHQ or 771-900mNQ
- [ ] c. 387-465mHQ or 576-700mNQ
- [ ] d. 465-551mHQ or 650-771mNQ

14. At what depth should you Preload and why is it important to preload when pulling and lowering rods?

- [ ] a. 500m HQ/600m NQ to eliminate shock loading the entire system
- [ ] b. 600m HQ/900m NQ to eliminate shock loading the foot clamp
- [ ] c. 700m HQ/900 NQ to eliminate shock loading the chuck
- [ ] d. 500m HQ/700m NQ to eliminate shock loading the entire system

15. Explain deep hole rod stretch.

- [ ] a. The small amount of "play" in each joint that culminates in a measurable amount at greater depth
- [ ] b. The amount steel stretches per meter of rod string
- [ ] c. The amount steel sags per meter of rod string
- [ ] d. The amount steel compresses under load

16. Which of the following is not a contributing factor to high water pressure that can affect your ability to drill?

- [ ] a. Excess rod grease
- [ ] b. Hole depth
- [ ] c. Clay seams
- [ ] d. Fractures

17. What is the max RPM that a B-20 head should go while drilling NQ?

- [ ] a. 1200 RPM
- [ ] b. 1250 RPM
- [ ] c. 1300 RPM
- [ ] d. 1350 RPM

18. **You will encounter rocks, boulders, clay and sand. These will all cause a variety of solvable problems, such as keeping the casing straight when starting a flat hole (i.e.: -45 degree), maximizing the life of the shoe, building a stable bore hole and preventing hole deviation before the casing is set into bedrock. Please provide a general description, based on your experience, explaining the method you would use to install casing?**

19. **What takes place when the tube latches and shuts off water?**

- [ ] a. The ball passes through the surelock into the lower latch body, creating a hydrostatic lock
- [ ] b. The back end compresses, causing the shut off valves to swell
- [ ] c. The compression spring extends, causing the lifter case to contact the crown of the bit
- [ ] d. All of the above

20. **What order, from softest to hardest, are the shut off valves?**

- [ ] a. Yellow, White, Black
- [ ] b. Black, Yellow, White
- [ ] c. White, Yellow, Black
- [ ] d. They're all fairly similar

21. **How do you measure Bit Weight with 1 cylinder?**

- [ ] a. Off Bottom Holdback Pressure – Drilling Holdback Pressure x 7 in2
- [ ] b. Off Bottom Holdback Pressure - Drilling Holdback Pressure
- [ ] c. Off Bottom Holdback Pressure - Drilling Holdback Pressure x 11 in2
- [ ] d. Off Bottom Holdback Pressure - Drilling Holdback Pressure x 9 ft2

22. **How do you measure Bit Weight with the Assist Cylinder?**

- [ ] a. Off Bottom Holdback Pressure – Drilling Holdback Pressure x 11 ft2
- [ ] b. Drilling Holdback Pressure - Off Bottom Holdback Pressure x 7 in2
- [ ] c. Off Bottom Holdback Pressure - Drilling Holdback Pressure
- [ ] d. Off Bottom Holdback Pressure - Drilling Holdback Pressure x 11 in2

23. **How do you measure how much flow is coming from the pressure pump in the field?**

- [ ] a. Read the manufacturer specs for the pump being used
- [ ] b. Do a hydro test (PSI x time to fill divided by volume)
- [ ] c. Do a bucket test (5 gal divided by time to fill x 60)
- [ ] d. Speed of rotation x volume of cylinders x 60

Cementing

For Questions 1-4, please refer to the information in appendix A.

1. **Calculate the required volume of cement to fill 150m of an HQ hole.**

   - [ ] a. 238 litres
   - [ ] b. 1011 litres
   - [ ] c. 1086 litres
   - [ ] d. 238 gallons

2. **Referring to question #1, how many barrels (total) are required?**

   - [ ] a. 3 barrels
   - [ ] b. 10 barrels
   - [ ] c. 6 barrels
   - [ ] d. 5 barrels

3. **Assuming 2-barrel batches, how many rods are pulled in total?**

   - [ ] a. 10
   - [ ] b. 17
   - [ ] c. 5
   - [ ] d. 20

4. **How many kilograms of cement are required per batch, and how many kilograms in total?**

   - [ ] a. 160 kg per batch, 480 kg total
   - [ ] b. 80 kg per batch, 480 kg total
   - [ ] c. 160 kg per batch, 960 kg total
   - [ ] d. 80 kg per batch, 960 kg total

5. **How many displacement plugs are used to pump a batch of cement to the bottom of the hole? Where do they go?**

   - [ ] a. 3 Plugs – 2 above cement, 1 below
   - [ ] b. 2 Plugs – 1 below, 1 above
   - [ ] c. 2 Plugs – 2 above cement
   - [ ] d. 3 Plugs – 2 below cement, 1 above cement

6. **You can pump cement through the water tree?**

   - [ ] a. True
   - [ ] b. False

7. **How far from the bottom do you lift the rod string when pumping a batch of cement to the bottom?**

   - [ ] a. 1-2 inches
   - [ ] b. 2-6 inches
   - [ ] c. 1-2 cm
   - [ ] d. 2-6 cm

8. **Referring to question #7, why do you lift this far off from the bottom?**

- [ ] a. So that you get an indication when the plugs reach bottom
- [ ] b. So that the cement doesn't go up your anulus
- [ ] c. So that your chase water doesn't dilute your cement
- [ ] d. So that you don't pressurize your hole

9. **What can happen if you pump cement from the bottom into the annulus (between the borehole and the rod string)?**

- [ ] a. Your cement sets up properly
- [ ] b. You can stick the rods and not have a proper column of cement
- [ ] c. You can pressurize the formation
- [ ] d. You'll have voids in your cement

10. **When pumping multiple batches of cement to cement an entire hole, why is it important to time the first batch and every batch following the first?**

- [ ] a. So that you know that will be the maximum time needed for the batch to be in place
- [ ] b. So that you know that will be the minimum time needed for the batch to be in place
- [ ] c. So that you know how much water is needed for each batch
- [ ] d. So that you know how much total volume is required for each batch

11. **How do you know when your second batch of cement has reached the top of the first batch?**

- [ ] a. Your water pressure gauge will reach 1000 psi and the 435 will stall
- [ ] b. There will be no suction from the open rod and the chase water will be at the top
- [ ] c. The water from the casing will stop and your water pressure gauge will reach 500 psi
- [ ] d. You will see your pressure spike when the plug clears the rods

Environment, Health & Safety

1. **Bentonite and cement products can cause several heath issues if not handled properly. Name the condition that it can cause and required PPE to be used when handling these products?**

- [ ] a. 3rd degree burns, elbow length gloves and goggles
- [ ] b. Cancer, N95 mask
- [ ] c. Severe eye irritation, safety glasses
- [ ] d. Silicosis, N95 mask

2. **When should you replace a hard hat?**

- [ ] a. If it shows signs of damage
- [ ] b. If it is dirty and greasy
- [ ] c. 5 years from manufacturer's date
- [ ] d. 3 years from the time you received it
- [ ] e. If you've spray painted it
- [ ] f. A & C
- [ ] g. A & D
- [ ] h. A, C & E

3. **Where can you find the manufacture date on a hard hat and for how many years is that hard hat good for?**

- [ ] a. Inside of hard hat brim and 5 years from manufacturer's date
- [ ] b. Back of hard hat and 4 years from manufacturer's date
- [ ] c. Inside of hard hat brim and 3 years from when you start wearing it
- [ ] d. Inside of hard hat and 5 years from when you start wearing it

4. **Why do you feel it is important to maintain eye contact if you are operating equipment or if you are working around an equipment operator?**

- [ ] a. The Operator is most important person on site and needs to be treated as such
- [ ] b. Operators are untrustworthy, they need to be watched at all times
- [ ] c. So that the Operator knows where you are at all times, and they can give you hand signals
- [ ] d. So that the Operator knows where you are at all times, and you can give them hand signals

5. **Before lifting and leveling a drill shack with a cat blade, a chain should be hooked up to the drill skid. Why should this action be completed before the cat is in position of lifting?**

- [ ] a. So that the person hooking up the skid to the blade isn't in the line of fire
- [ ] b. To make it faster and more efficient
- [ ] c. To not get the chain muddy
- [ ] d. Use the hydraulic jacks on the drill instead

6. **What is the signal for 'down' when receiving a helicopter fly load?**

- [ ] a. Big circles above your head
- [ ] b. Back and forth chopping motion above your head
- [ ] c. Arms extended out to your sides
- [ ] d. One arm extended with your thumb pointing down to the ground.

7. **What does this hand signal mean?**



- [ ] a. Load up
- [ ] b. Load Down
- [ ] c. Place Load Here
- [ ] d. I need a hug

8. **What does this hand signal mean?**



- [ ] a. Hold
- [ ] b. Stop
- [ ] c. Abort
- [ ] d. Down

9. **What is the proper way to fuel up equipment?**

- [ ] a. Use a secondary containment, have a spill kit available, place the nozzle in and use the lock until it clicks off
- [ ] b. Use a secondary containment, have a spill kit available, shut off equipment and fuel up while having a smoke
- [ ] c. Use secondary containment, have a spill kit and fire extinguisher available, shut off equipment and stay present while fueling
- [ ] d. Use secondary containment, have a spill kit and fire extinguisher available, shut off equipment and do your pre-op while it fuels up

10. **When using a fire extinguisher, what does the acronym P.A.S.S. mean?**

- [ ] a. Pull, Aim, Squeeze, Sweep
- [ ] b. Pin, Aim, Squeeze, Sweep
- [ ] c. Pull, Aim, Sweep, Squeeze
- [ ] d. Pull, Act, Squeeze, Sweep

11. **What are the steps to lock out a piece of equipment?**

- [ ] a. Correct all gravity hazards, shut down, release all stored energy, de-energize machine, attach hasp and lock
- [ ] b. Correct all gravity hazards, shut down, release all stored energy, de-energize machine, attach hasp and lock, try to start the machine
- [ ] c. Correct all gravity hazards, shut down, de-energize machine, release all stored energy, attach hasp and lock, try to start machine
- [ ] d. Correct all gravity hazards, shut down, de-energize machine, attach lock and hasp, release store energy, try to start machine

12. **In which of the following scenarios would you utilize FO-100 Safety - Job Hazard Analysis Tool (JHA) and why?**

- [ ] a. When doing a high-risk task, because there may be hazards that employees are not aware of
- [ ] b. When doing something for the first time, so that everyone understands the job
- [ ] c. When your Forman tells you to, because he's the boss
- [ ] d. A & B

13. **What step needs to be done first before doing a live lock out?**

- [ ] a. Place signage at the control panel
- [ ] b. Ensure there is a guard in place so that no one can engage any functions
- [ ] c. Print off and complete a FO-299 Live Lock Out form
- [ ] d. Complete a JHA with all individuals involved

14. **How often should crew members be encouraged to stretch during a shift?**

- [ ] a. In the morning before the start of shift
- [ ] b. Once every hour
- [ ] c. Once every 6 hours
- [ ] d. Only as needed throughout the shift

Mud Tank

1. **The foreman is responsible for ensuring the driller and helper fully understand the process involved when a mud program is needed for difficult ground/drilling conditions. Who is responsible for ensuring the consistency of the drill mud at the drill according to the program?**

- [ ] a. The Client
- [ ] b. The Foreman
- [ ] c. The Helper
- [ ] d. The Driller

2. **Explain the driller's responsibility when maintaining mud consistency.**

- [ ] a. To show the helper how to properly mix mud and check it periodically throughout the shift to make sure it is the right viscosity and not too much is being used
- [ ] b. To train the Helper to mix mud, once he has done that, it is the responsibility of the helper to check it periodically with a vis cup to make sure the right amount is being used
- [ ] c. To mix it himself so that the viscosity is right and that not too much is being used
- [ ] d. To make sure that there is enough mud in the hole to keep the torque down while drilling

3. **What is the purpose of Bentonite?**

- [ ] a. Bentonite is a silicate clay mud that builds a wall in the drill hole to stabilize the formation and prohibit drilling solution from permeating the formation.
- [ ] b. Bentonite is a silicate clay mud that flushes the hole and carries cuttings to the surface.
- [ ] c. Bentonite is a polymer mud that builds a wall in the drill hole to stabilize the formation and prohibit drilling solution from permeating the formation.
- [ ] d. Bentonite is a polymer thickener that adds viscosity to the drilling solution.

4. **How do you measure the viscosity of drilling fluid?**

- [ ] a. Use a mud funnel, fill to the line with finger on the bottom and time how long it takes to empty
- [ ] b. Use a mud funnel, fill to the line through the screen attached to the funnel with gloved finger on bottom opening, measure how long it takes to empty the funnel
- [ ] c. Use a mud funnel, fill to the line through the screen attached to the funnel with gloved finger on bottom opening, measure the time it takes to fill to the bottom line of mud cup
- [ ] d. Use a mud funnel, fill to the line through the screen attached to the funnel with gloved finger on bottom opening, measure the time it takes to fill to the top line of mud cup

5. **What is water pH? What pH number is neutral and what number is best for mixing mud?**

- [ ] a. Neutral is 7pH. Best mud mixing occurs between 8-9pH
- [ ] b. Neutral is 5pH. Best mud mixing occurs between 6-7pH
- [ ] c. Neutral is 6pH. Best mud mixing occurs between 7-8pH
- [ ] d. Neutral is 7pH. Best mud mixing occurs between 6-7pH

6. **What happens if bentonite is mixed after other drilling additives?**

- [ ] a. It will harden like cement
- [ ] b. It will mix properly
- [ ] c. It will separate downhole
- [ ] d. It will ball up and cause trouble pumping

7. **What do we use to change the pH of our water to help with mixing mud?**

- [ ] a. Caustic lime
- [ ] b. Soda ash
- [ ] c. Sodium hydroxide
- [ ] d. Baking soda

8. **What is the purpose of a clay stabilizer and what are the benefits of using this product?**

- [ ] a. A clay stabilizer breaks down the clay through decomposition so that you can drill through it
- [ ] b. A clay stabilizer helps with hydration of the clay with oppositely charged ions that bond to the clay formation, preventing it from swelling
- [ ] c. A clay stabilizer inhibits hydration of the clay with oppositely charged ions that bond to the clay formation, preventing it from swelling
- [ ] d. A clay stabilizer seals the clay so that water migration cannot occur and create swelling

9. **What type of mud is 650?**

- [ ] a. a collector
- [ ] b. an inhibitor
- [ ] c. a polymer
- [ ] d. a stabilizer

Underground

1. **When pulling rods on an up hole, what precautions should be taken in order to ensure the safety of workers?**

2. **What is the Cap Lamp signal for « come to me »?**

   - [ ] a. Side to side (NO)
   - [ ] b. Up and down (Yes)
   - [ ] c. Big Circle

3. **What is the cap lamp signal for « retreat from area »?**

   - [ ] a. Side to side (NO)
   - [ ] b. Up and down (Yes)
   - [ ] c. Big circle

4. **What is the cap lamp signal for « stop there »?**

   - [ ] a. Side to side (NO)
   - [ ] b. Up and down (Yes)
   - [ ] c. Big circle

5. **What is one of the hazards of pulling a tube on an up hole and what should you do to avoid this?**

   - [ ] a. The rod string might slip back, stand clear until the driller gives the OK
   - [ ] b. There might be a piece of core that comes out, stay clear and listen for ringing before loading the next tube
   - [ ] c. The tube could come out of the hole unexpectedly, stand clear and catch it as it comes out
   - [ ] d. The overshot might not be latched, be ready to throw it back in when it comes out

6. **When should you use a block up plug?**

   - [ ] a. When you have more than 15 gpm
   - [ ] b. When you're doing a cement job
   - [ ] c. End of hole to "cap" hole.
   - [ ] d. When you have a stuck tube and need to pull

7. **How do you install a block up plug?**

   - [ ] a. Inspect plug, preload plug into rod with swivel, install rod, pump at medium speed until pressure comes up, shut water off, pump again with less water 2-3 times so that air is bled through.
   - [ ] b. Inspect plug, preload plug into rod with swivel, install rod, pump at max speed until pressure comes up, shut water off, pump again with less water 2-3 times so that air is bled through
   - [ ] c. Inspect plug, load plug into rod string, install swivel, pump at low speed until pressure comes up, shut water off, pump again with less water 2-3 times so that air is bled through
   - [ ] d. Inspect plug, preload plug into rod with swivel, install rod, pump at medium speed until pressure comes up, shut water off, if water pressured up pull rods

8. **What gasses does the Altair 4 test for?**

- [ ] a. Methane, Propane, Carbon monoxide, Oxides of Nitrogen
- [ ] b. Oxygen, H2S, Oxides of Nitrogen, Propane
- [ ] c. Helium, Methane, Propane, Oxygen
- [ ] d. Oxygen, Methane, Oxides of Nitrogen, Carbon Monoxide

9. **At what % or how many PPM of explosives (Ex/Comb) including methane, do electric motors need to be stopped?**

- [ ] a. 1.25% or 12500ppm
- [ ] b. 2.5% or 25000ppm
- [ ] c. 0.125% or 1250ppm
- [ ] d. 0.25% or 2500ppm

10. **What is the minimum % of oxygen required to work underground?**

- [ ] a. Greater than 18%
- [ ] b. Less than 19.5%
- [ ] c. Greater than 16.5%
- [ ] d. Greater than 19.5%

11. **Threshold Limit Values (TLV) refer to airborne concentrations of substances and represent conditions under which it is believed that nearly all workers may be repeatedly exposed day after day without adverse effect. (TLV/TWA): the time-weighted average concentration to which nearly all workers may be repeatedly exposed, day after day, without adverse effects is based on which time model?**

- [ ] a. 12-hour day / 84-hour week
- [ ] b. 8-hour day / 40-hour week
- [ ] c. 10-hour day / 40-hour week
- [ ] d. 10-hour day / 70-hour week

12. **What is the TVL-STEL of Carbon Monoxide?**

- [ ] a. 100 ppm
- [ ] b. 50 ppm
- [ ] c. 25 ppm
- [ ] d. 15 ppm

13. **What is the TVL (Threshold Limit Value) of Oxides of Nitrogen?**

- [ ] a. 0.0003% (3ppm) maximum exposure at any time or 0.0001% (1ppm) maximum exposure at any time (BC)
- [ ] b. 0.0004% (4ppm) maximum exposure at any time or 0.0001% (1ppm) maximum exposure at any time (BC)
- [ ] c. 0.0003% (3ppm) maximum exposure at any time or 0.0002% (2ppm) maximum exposure at any time (BC)
- [ ] d. 0.0006% (6ppm) maximum exposure at any time or 0.0003% (3ppm) maximum exposure at any time (BC)

14. **When do you use a self rescuer?**

- [ ] a. When there is no Oxygen
- [ ] b. When you smell stench gas
- [ ] c. On the way to the refuge
- [ ] d. When you think there is a fire or explosion underground

15. What reaction happens in a W-65 that enables you to continue breathing?

- [ ] a. Converts Carbon Monoxide into Oxygen
- [ ] b. Filters out Carbon Dioxide
- [ ] c. Converts Carbon Monoxide into Carbon Dioxide
- [ ] d. Filters out Carbon Monoxide and Carbon Dioxide

16. Why do you chain a Margo plug to the face?

- [ ] a. To secure it in case air pressure pushes it out
- [ ] b. To secure the screen around the plug
- [ ] c. So that you can anchor the drill to it
- [ ] d. So that the plug is secured in case water pressure pushes it out

17. What are the differences between uphole backends and downhole backends?

- [ ] a. The ears
- [ ] b. The links
- [ ] c. The hanger bearings
- [ ] d. The shutoff rubbers

18. A downhole upper latch body can be installed on an uphole backend?

- [ ] a. True
- [ ] b. False

19. How many scaling bars are required at a drill station?

- [ ] a. 3 Bars. 1 long, 1 medium, 1 short.
- [ ] b. 4 Bars. 2 short, 1 medium, 1 long.
- [ ] c. 1 Bar. 1 short.
- [ ] d. 2 Bars. 1 long, 1 short.

20. Where do the scaling bars need to be stored?

- [ ] a. At the Storage
- [ ] b. At the Drill
- [ ] c. At the Entrance
- [ ] d. At the Re-muck

21. Select the answer to best explain proper scaling procedure?

- [ ] a. Choose the correct length scaling bar, work from good ground to bad ground. Maintain a clear path of escape. Maintain proper footing. Stand at a 45° angle to the loose being removed with the scaling bar positioned along side of your hip. If the loose is too high or too large to scale, the area must be roped off to prevent unauthorized entry and the supervisor or shifter must be notified immediately.
- [ ] b. Work from good ground to bad ground. Maintain a clear path of escape. Stand at a 45° angle to the loose being removed with the scaling bar positioned along side of your hip. If the loose is too high or too large to scale, the area must be roped off to prevent unauthorized entry and the supervisor or shifter must be notified immediately.
- [ ] c. Bang away at it until it falls off the wall.
- [ ] d. Choose the correct length scaling bar. Work from good ground to bad ground. Maintain a clear path of escape. Maintain proper footing. Stand at a 75° angle to the loose being removed with the scaling bar positioned along side of your hip. If the loose is too high or too large to scale, leave it for your cross shift.

22. Where do you install the jackleg lubricator and how long should the hose be?

- [ ] a. On the jackleg, 5′ hose
- [ ] b. Between the jackleg and the header, 20′ hose
- [ ] c. Hung on the rib, 40′ hose
- [ ] d. On the rod rack, 20′ hose

23. Are you allowed to tag in or out for someone else?

- [ ] a. Yes, if you have verbal permission from the person
- [ ] b. No, it is never acceptable to remove someone's tag
- [ ] c. Yes, if the Mine Manager removes the tag
- [ ] d. Yes, if you have written permission to remove the tag

24. When do you perform a visitor orientation?

- [ ] a. Every time someone comes to your drill
- [ ] b. When someone comes to your drill, unless they were at your drill on the last pad
- [ ] c. When a person that is not a Hy-Tech employee comes to your drill
- [ ] d. When someone is visiting and they have not been to this pad before

25. What is stench gas?

- [ ] a. An emergency alert system letting mine workers know to head to safety
- [ ] b. A method of letting mine personnel know that the shift is over
- [ ] c. A warning system that there is a fire underground
- [ ] d. A method of circumventing radio communication

26. What do you do when you smell stench gas?

- [ ] a. Drive to surface
- [ ] b. Keep drilling until a supervisor tells you to stop
- [ ] c. Call on the radio to see if there is an emergency
- [ ] d. Head to the nearest refuge and inform supervisor

27. Why do workers put up barricades and what should the signage indicate?

- [ ] a. To prevent unwanted visitors to come on the pad, channel that drill crew can be reached on
- [ ] b. To prevent potential danger to someone in the mine, reason for barricade and who to call for access
- [ ] c. That work is being conducted in the area and the phone number to call for access
- [ ] d. High CO and do not enter

28. When collaring a hole, how do you prevent core barrel deflection when the core barrel is going to contact the face at an angle?

- [ ] a. Use high rotation and light penetration with a standard barrel
- [ ] b. Use low rotation and high penetration with a standard barrel until a lip is formed
- [ ] c. Hold it in place with a bar as you do the first cut, take small strokes with chuck
- [ ] d. Use a 1.5m barrel with low rotation and light penetration and take small strokes with chuck

29. What is the minimum distance needed to collar a hole beside a bootleg or lifter?

- [ ] a. 4″
- [ ] b. 6″
- [ ] c. 8″
- [ ] d. 12″

30. What is the minimum distance needed to collar a hole next to a cut?

- [ ] a. 6''
- [ ] b. 10''
- [ ] c. 12''
- [ ] d. 18''

31. What do you do if you can't latch your tube on an uphole?

- [ ] a. Pull rods
- [ ] b. Bail rods until it latches
- [ ] c. Follow block up plug procedure and pull rods
- [ ] d. Drill 6'' and try to break core

32. If your tube is stuck on an uphole, what do you do?

- [ ] a. Feed pressure into the bottom of the hole so that it pushes the tube out of the core barrel
- [ ] b. Pull the wireline in short jerks to loosen the tube out
- [ ] c. Bail the rods up and down to free tube
- [ ] d. Send a block up plug up and pull rods

33. How do you properly set your tube to shut off water on an uphole?

- [ ] a. Loosen the compression spring continually until the tubes shut off water
- [ ] b. Set the tubes so that they are ⅛-¼**''** from the bit and the rubbers can be barely turned by hand
- [ ] c. Replace stiffer rubbers with softer rubbers
- [ ] d. Put a high-pressure landing indicator bushing in the lower latch body

34. If the ventilation system fails or shuts down, what must you do and why?

- [ ] a. Investigate the cause, call the Foreman, after 15 min if ventilation doesn't return move to a ventilated area and inform the Foreman of new location
- [ ] b. Call the Foreman, after 20 min if ventilation doesn't return move to ventilated area and inform Foreman of new location
- [ ] c. Investigate the cause, call Mine Captain, after 25 min if ventilation doesn't return move to a ventilated area and inform Mine Captain of new location
- [ ] d. Keep drilling, ventilation will fix itself eventually

Why?

35. What gear should you be in while driving the Toyota down ramp?

- [ ] a. 1st
- [ ] b. 2nd
- [ ] c. 3rd
- [ ] d. 4th

Supervisor

1. **Who is responsible for training New and Young Workers?**
   - [ ] a. Foreman
   - [ ] b. Trainer
   - [ ] c. Fieldman
   - [ ] d. Driller
2. **Give 2 examples of a New Worker?**
   - [ ] a. New to industry, under 20 years old
   - [ ] b. New to industry, under 25 years old
   - [ ] c. New to industry, under 18 years old
   - [ ] d. New to industry, under 22 years old
3. **How often does a Foreman conduct a drill site audit?**
   - [ ] a. Once a week
   - [ ] b. Every 5 days
   - [ ] c. On safety Sundays
   - [ ] d. When he has time
4. **When do you begin the 5-point section in your time card?**
   - [ ] a. When you get to the drill
   - [ ] b. Mid-shift when you've had time to assess the conditions
   - [ ] c. At the end of your shift, so that your time card stays clean
   - [ ] d. At the beginning of your shift, before any work commences
5. **What is the first point in the 5-point section?**
   - [ ] a. Is equipment in working order
   - [ ] b. Check entrances and travel ways
   - [ ] c. Do an act of safety
   - [ ] d. Writing your name and drill number
6. **How often should a pre-operational check be completed?**
   - [ ] a. Before every use
   - [ ] b. At the beginning of your shift, or before starting a piece of equipment
   - [ ] c. During the shift, once the equipment has warmed up
   - [ ] d. At the end of shift, to make sure it's good for your cross shift
7. **Where can you find the FO-074A Safety - Orientation and Task Checklist; Driller's Helper?**
   - [ ] a. In the red toolbox
   - [ ] b. Behind the driller on the wall
   - [ ] c. In the Foreman's office
   - [ ] d. In the Red Envelope

8. **Where can you find the FO-100 Safety – Job Hazard Analysis Tool (JHA)?**

- [ ] a. The Foreman's office
- [ ] b. The red job box at the drill
- [ ] c. The OH&S rep on site
- [ ] d. In the Hy-Tech app on the tablet

9. **What is the significance of the Westray Bill C-45? To stress the responsibility taken on by the driller at the worksite every day. There are legal ramifications for negligence on the part of the Driller should anything happen to the helper or anyone else on the worksite under the supervision of the Driller.**

10. **Who can you bring concerns to regarding workplace health and safety?**

- [ ] a. Foreman
- [ ] b. Drillers
- [ ] c. OHSC Members
- [ ] d. All of the above

Appendix A

SIZE	HOLE VOLUME (L/M)	HOLE VOLUME (IMP G/M)	# M/BATCH OF CEMENT (180L OR 40G)	# RODS/BATCH OF CEMENT (180L OR 40 G)
BQ - 3 M ROD	2.82 L/M	0.62 G/M	63.82 M	21.3 RODS
NQ - 3 M ROD	4.51 L/M	1.00 G/M	39.91 M	13.3 RODS
HQ (NW) – 3 M ROD	7.24 L/M	1.59 G/M	24.86 M	8.3 RODS
PQ (HWT) – 1.5 M ROD	11.80 L/M	2.60 G/M	15.25 M	10.2 RODS

NOTE: Most of the drums that are currently being used for cement mixing are 45-gallon drums (203 litres) when filled to within 2 cm of the top. We typically will fill a drum to the second ring (about 140 litres of water), and add 160 kg of Portland type 10 cement, to get approximately 180 litres per batch when completed.

Please provide your feedback in the space below and give any suggestions that you feel would improve the process.